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Original Article

Failure analysis on silicon semiconductor device materials: optical and high-resolution microscopic assessments



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ABSTRACT

Defects of silicon (Si) semiconductor epilayers are crucial to be identified at laboratory environs. The identification of failure and its rectification at laboratory settings is essential for large-scaling manufacturing of narrowed down semiconductor devices. This research documented the inspection, identification and the solution for defects found in the Si semiconductor epilayers, fabricated by a simple and conventional photolithography technique, with the integration of metal oxide nanomaterial, zinc oxide (ZnO). The semiconductor epilayers, Si wafer, Si oxide and ZnO coated SiO₂ layer were formed and examined. Optical microscope images [high power microscope (HPM) and 3D profilometer] reveal smooth surface of semiconductor epilayers development through thermal oxidation and photolithography techniques. High power ultraviolet-visible (UV-Vis) justified the accuracy of wet thermal oxidation by examining the thickness of oxide layer on Si wafer at 3837.3 Å. The X-ray diffraction (XRD) analysis of sol-gel synthesized ZnO affirmed the

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E-mail addresses: subash@unimap.edu.my (S.C.B. Gopinath), md_ibrahim83@utm.my (M.I. Shapiai Razak).¹ Equally contributed.<https://doi.org/10.1016/j.jmrt.2022.10.116>2238-7854/© 2022 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

hexagonal crystalline state and its nanoscale size at 54 nm. Field emission scanning electron microscopy (FESEM) has shown the insight of Si epilayer morphology with its elemental composition, which provides details of foreign substances on semiconductor surface. ZnO deposited Si epilayer was prepared through lamella preparation, prior to the cross-sectional field emission transmission electron microscopy (FETEM) analysis of the semiconductor, which revealed the uniformity of fabrication and ZnO distribution at Si epilayer. Failure analysis reported several defects on the Si epilayers in the state of patches and accumulation of impurities. The potential cause of the defects and the respective solutions are discussed as the accuracy and handling must be ensured throughout the fabrication process, to develop a flawless semiconductor for high performance applications.

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1. Introduction

Semiconductors are in demand to strengthen the science and engineering technology forward into a new era. Semiconductors are utilized from the heavy to light industries such as, automotive, manufacturing, telecommunications and computers, for their efficiency in profit with very low energy loss system [1]. The on-going studies are mainly attentive in semiconductor epilayers and their dimensions to develop a power pack semiconductor chip to advance the usage in economic scales. The manipulations in semiconductor epilayers in term of materials, preparations, doping, etching and thin films have received a huge attention in developing high performance semiconductors [2,3]. Materials are one of the main criteria in developing a potential semiconductor. Silicon (Si) based semiconductors are the primitive types of chips that are widely applied and upholds their position at the first place in the large-scale industries. Si wafer is the primary substrate, which also categorized as individual semiconductor. It serves as the thin slide of semiconductor [4,5]. Si wafer easily recognized for its dominant use in fabrication of integrated circuit (IC) and photovoltaic (solar cells). Si wafers are commonly used in photolithography patterning, thin-film deposition, doping and ion implantation, etching and many more fabrication processes. Si is considered as a worthwhile material due to its naturally exhibiting low thermal conductivity and relatively high electrical conductivity with good band gap and its substantial availability with low cost [6,7]. Si oxide thin layer used as dielectric material and integrated with micro electrochemical system (MEMS) devices. It is a semiconductor, serves as an intermediate medium for device development and processing according to the desired operation. Wet and dry thermal oxidations of Si wafer are common to grow the Si oxide thin layer. Thermal oxidation often performed at 850–1200 °C. Thermal oxidation indicate low oxide growth rates. It allows thickness uniformity and purity [8].

Multi-junction semiconductors are constructed with the addition of materials (nanoscale) as thin films to enhance the semiconductor electric conductivity and energy absorptions [9]. Transition metals and their oxides are the most promising thin film materials with narrow band gap and good infrared

absorption capacity [10]. Zinc oxide (ZnO) nanomaterial is one of commonly recognized metal oxide semiconductors in microfabrication. High sensitivity and ZnO is an ideal choice due to its wide energy band gap with 3.37 eV and high bond energy with 60 meV at room temperature [11,12]. In addition, ZnO is an attractive material applicable for immobilization of biomolecules without an electron mediator due to its non-toxicity, high chemical stability and high electron transfer potential [13,14]. The stacked ZnO with other metal oxides enables good conformity and uniformity for producing high performance transistors. The combination of ZnO with other dielectric oxides at a multilayered heterostructure is highly encouraged in transistors and semiconductors. The engineering of conductive and insulative oxide layers facilitates the orientation of a component and controls the defect passivation. The stacked ZnO layer able to withstand the bias pressure, especially in developing thin film transistors. Moreover, multilayered ZnO is in demand for developing quantum dot-light emitting diodes, due to the good electron mobility at the transport layer and the good thermal stability [15,16]. ZnO are well adapted in the preliminary testing of semiconductor performances with the integration of nanoscale materials. It is preferred for its easy availability and cost-effective synthesizing methods [17].

The future envisions of electronic industry necessitate semiconductors scale-down, in term of its morphology and features for its utmost performances and satisfactory fitting to multiple applications. A mass scaling-down of semiconductors for highly reliable electronic gadgets with high manufacturing yield are ensured with high accuracy fabrication and sophisticated microscopic techniques [18]. Microscopic failure analysis is an inevitable step in the semiconductor fabrication. Semiconductor failure analysis determines the root cause of semiconductor limitations, in term of its material, dimension, parameters, and morphology, prior to the large-scale manufacturing of electronic chips and devices. It is a vital element of semiconductor fabrication for delivering reliable evaluation on the device quality and performance. The extensive cause of a semiconductor failure is classified into four main causes, which are the material fault, human error, machine malfunctions, and the mode of operation. Prior to identifying the cause of a failure or defect, it is important to determine the

mode of failure. The failure mode usually interpreted in device malfunction, such as short circuit, open circuit, leakages, and other poor electrical performances [19]. Besides, the preliminary failure analysis is initiated from defect analysis, where the device is inspected for cracks, impurity growth, micro-faults on its morphology. Examining devices in both defective and good condition is beneficial for failure analysis. If the study on induced failures can be carried out at the very beginning of device development, it can be of great assistance. It is essential to use a systemic, step-by-step method for failure analysis. However, the wide range of current technologies and gadgets necessitates a wide range of failure analysis methods [20]. Laboratory based failure analysis using microscopic techniques are notable among the existing modern defect inspection tools. The electron and optical imaging tools aid in the critical failure analysis of semiconductors and its epilayers for significant defect visualization from micro-to nano-scale dimensions. The advancement attained in electron and optical microscopic techniques enable an inexpensive and easy handling laboratory-based failure analysis to precisely inspect and resolve the deviations in semiconductors from its ideal state [21]. The presented work represents the laboratory-based Si semiconductor epilayers fabrication and its inspection through electron and optical microscopic techniques. ZnO nanomaterial is synthesized from sol-gel technique applied in the research to create a thin film on Si semiconductor. The defects associated with material and morphology at the multiple epilayers of fabricated Si semiconductors are determined. The presented research demonstrates the potential of simple and straightforward microscopic techniques for inspecting of semiconductor epilayers and perform critical failure analysis in growing semiconductor industry.

2. Methodology

2.1. Materials and reagents

The primary epilayer, Si wafers were procured from Malinkrodt Baker, Ireland. The analytical materials and solutions used in photolithography technique and ZnO sol-gel method, such as Al etch solution, ZnO precursor, isopropanol and ethanolamine were obtained from Sigma Aldrich, USA. Photoresist solutions and its developing agent were purchased from Futurrex Inc, USA.

2.2. Fabrication of semiconductor epilayers

Si wafer represent as the primitive epilayer of semiconductor. Wet thermal oxidation was performed to develop and thin oxide layer, as Si oxide on the substrate. The Si wafer was placed in wet thermal oxidation furnace at 500 °C for 1 h to generate a thin film of Si oxide epilayer with approximately ~300 Å thickness. The thickness was examined using high power ultraviolet-visible (UV-Vis) profiler (F20-UVX). The photolithography process of SiO₂ substrate was carried out as discussed in the previous works [18,22]. The epilayers were characterized using optical microscopes, which are high power microscope (HPM; Nikon Corporation, Japan) and 3D profiler (Hawk nano-profilometer) and electron microscopes, which are

field emission scanning electron microscope (FESEM, Hitachi, S-4300 SE, Japan) and field emission transmission electron microscope (FETEM, F20-Tecni, Columbus). The FESEM analysis were performed with the concept of backscattered electrons (BSE), where high-energy electrons are used to image the distribution of defect elements of semiconductor epilayers at high-resolution. The most common bright field (BF) mode was applied to analyze the defects at semiconductor epilayer using field emission transmission electron microscopy (FETEM). The chemical functional group present at each semiconductor epilayers were characterized using Fourier transform infrared spectroscopy (FTIR, Agilent, USA).

2.3. Preparation and deposition of ZnO

ZnO seed solution was prepared using sol-gel method, where zinc acetate dehydrate was used as the precursor and isopropanol as the solvent. A 0.2 M of zinc acetate dehydrate was prepared using 99% isopropanol solvent. The mixture was stirred at 1000 rpm and 65 °C for 30 min to completely dissolve the zinc precursor. Ethanolamine as stabilizer was prepared in equivalent ratio to zinc precursor. Then, ethanolamine was added into the mixture, drop by drop for every 2 h. The reaction mixture is maintained at stirring condition with 60 °C temperature. Once completed, the stirring speed is reduced, and the mixture kept overnight for aging before use. X-ray powder diffraction (XRD, D2 PHASER Bruker) analysis was performed to analyze the purity of the ZnO. The prepared ZnO seed solution was deposited on SiO₂ semiconductor epilayer using spin coating and drop casting techniques. The spin coating method was used by placing SiO₂ wafer on spin coater, set at 3000 rpm for 60 s. The ZnO solution was dropped and allowed to form a uniform thin film of ZnO. The wafer was annealed at 150 °C for 30 min. Then, ZnO spin coating process was repeated to develop a double layer of ZnO on the SiO₂ wafer. The variation between single and double layer of ZnO was analyzed. The drop casting method was conducted by adding ZnO seed solution on SiO₂ wafer and proceeded with annealing at 500 °C for 2 h. The ZnO thin film coated semiconductor epilayers were characterized using optical microscopes.

3. Results and discussion

The morphological characteristics of semiconductor epilayers were observed and analyzed through optical and electron microscopes. The defects identified in the semiconductor epilayers were discussed. The potential sources of the defects and the possible countermeasures are elaborated.

3.1. Silicon semiconductor epilayers: morphology and thickness analysis

Fig. S1 shows the HPM images of Si wafer and SiO₂ epilayers. Planar Si wafer (Fig. S1a) and Si oxide (Fig. S1b) thin layer has smooth surface as observed. The optical microscope image shows the smooth and clear surface of Si wafer and its thin oxide layer. Fig. S1c and S1d reveal the HPM images of SiO₂. The sharp edges of Al on SiO₂ epilayer justify the accuracy of

photolithography fabrication of the semiconductor. The Al pad aid in the electrical characterization of Si oxide semiconductor. SiO₂ epilayer referred as insulator, thus Al pad was fabricated in the epilayer to ensure the presence of conductive electrode for electrical analysis. The thickness of oxide layer on Si wafer after wet thermal oxidation was examined using high power UV-Vis. The reference Si wafer with 5000 Å was used for the analysis. Fig. S2 shows the UV-Vis profiler indicating the variation in oxide layer thickness. The analysis resulted that the actual SiO₂ epilayer thickness is 3837.3 Å, with 0.99 goodness of fit. The thickness of SiO₂ epilayer was analyzed using a nominal SiO₂ wafer as the standard sample for the test. The output of UV-Vis analyzer is the thickness and the goodness of fit, which indicates the exactness of sample from the standard nominal thickness. The goodness of fit at 1 indicates the actual sampling and standard sample has the same quality. If the value is vice-versa, the data is skewed from the standard sample [23,24]. The analysis justified the accuracy of wet thermal oxidation as the goodness of fit is close to 1, where the thickness of SiO₂ from wet thermal oxidation is close to the nominal SiO₂ wafer.

3.2. ZnO deposited SiO₂: HPM analysis

Fig. 1 shows the ZnO nanomaterial deposited in SiO₂ layer using spin coating technique. Fig. S3 shows the sol-gel synthesized ZnO nanomaterial and its XRD spectra. As shown in Fig. 1a, b and 1c, ZnO is not apparently visible for a single layer spin coating on SiO₂ due to the very thin layer of

nanomaterial. Nevertheless, a uniform ZnO deposition is apparent for double layers ZnO deposition, as shown in Fig. 1d, e and 1f. The defect density may increase with the increase deposition layers of ZnO due to the roughness and the individual grain size of ZnO. It adds the thickness of sensor surface and causes overall modification on the sensor morphology. Studies have reported the defect points sol-gel synthesized ZnO nanomaterial as single or double layer on a semiconductor. The performance of the nanomaterials is tuned as per the desired outcome in term of its conductivity. By controlling the density of defect points, and nature of impurities, the ZnO properties are tuned. Hence, both single and double layered ZnO are utilized for developing high-performance semiconductors [25]. The optical HPM unable to indicate the variation in defect density for the multiple ZnO deposition layers. However, the 3D profilometer optical microscope has showed the extend of surface roughness with single and double layer ZnO deposition (Section 3.3). Deposition of ZnO nanomaterial using ZnO seed solution was obtained from sol-gel technique, highly prefers more than single layer deposition to ensure a thicker seed layer, which enhances the well-aligned nanomaterial film for further electrical modifications [26]. Fig. 1g, h and 1i show that the ZnO drop-casting on SiO₂ epilayer the nanomaterial deposition is not uniform on SiO₂ surface. ZnO particles aggregated, and do not perfectly deposit on SiO₂ surface. Hence, spin coating method is the preferred over drop casting to deposit metal oxide nanomaterial on Si semiconductor epilayer.

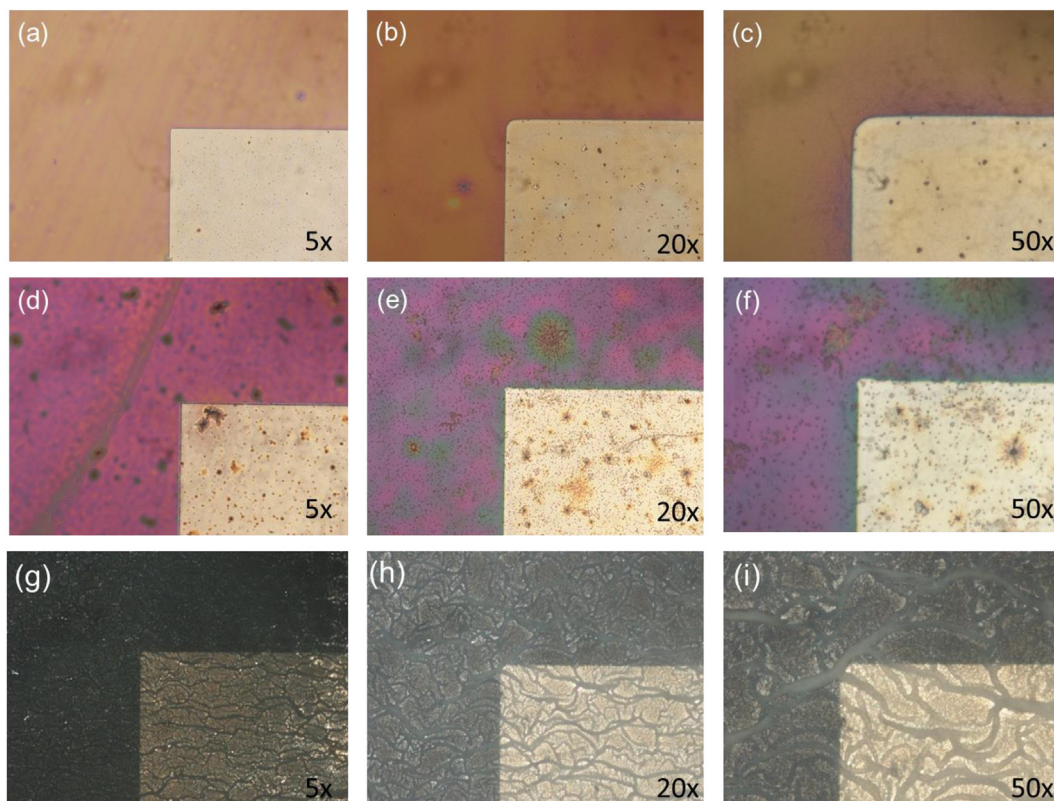


Fig. 1 – HPM images of ZnO deposited SiO₂ epilayer. (a)–(c) Single layer of ZnO spin coating deposition on SiO₂. (d)–(f) Double layer of ZnO deposited on SiO₂ using spin coating. (g)–(i) ZnO drop casting images of ZnO on SiO₂.

3.3. 3D profiler analysis

Fig. 2 shows the 3D images and their surface roughness plots obtained from 3D profiler. Fig. 2a shows the smooth surface of Si wafer. The 3D image justifies almost no impurities on substrate. The surface roughness is very low as surface profile shows very low variation in thickness. Fig. 2b represents the 3D of oxidized Si wafer after thermal wet oxidation, which results in smooth Si oxide thin layer. The surface profile indicates the consistency of its smoothness, grounds the accuracy of wet thermal oxidation process of Si wafer. The smooth surface of fabricated SiO_2 surface is shown in Fig. 2c. The 3D image shows the presence of Al electrode pad region with the appearance of an upper and lower surfaces, refer to Al pad and SiO_2 , respectively. It is further affirmed by the surface profiler graph, showing the higher and lower

roughness curve in the plot. It indicates the thickness difference between SiO_2 and Al electrode. Fig. 2d represents the 3D image of SiO_2 surface smoothness after single layer ZnO coating. The surface profiler shows that ZnO coating increase the roughness of SiO_2 surface. The presence of ZnO nano-materials is clearly observed in Fig. 2e, where double layer ZnO spin coating was performed. The surface roughness shown in surface profile indicate double layers of ZnO coating has reduced the smoothness due to the slightly thick layer of the nanomaterial. Moreover, spin coating method has given a uniform deposition of ZnO on SiO_2 surface, proven by the least variation shown in the surface profiler. The ZnO drop cast on SiO_2 is not uniformly dispersed and formed clumps of particles, as shown in Fig. 2f. The particle aggregation creates high surface roughness as seen in surface profile. Based on the 3D profiler analysis, the Si wafer,

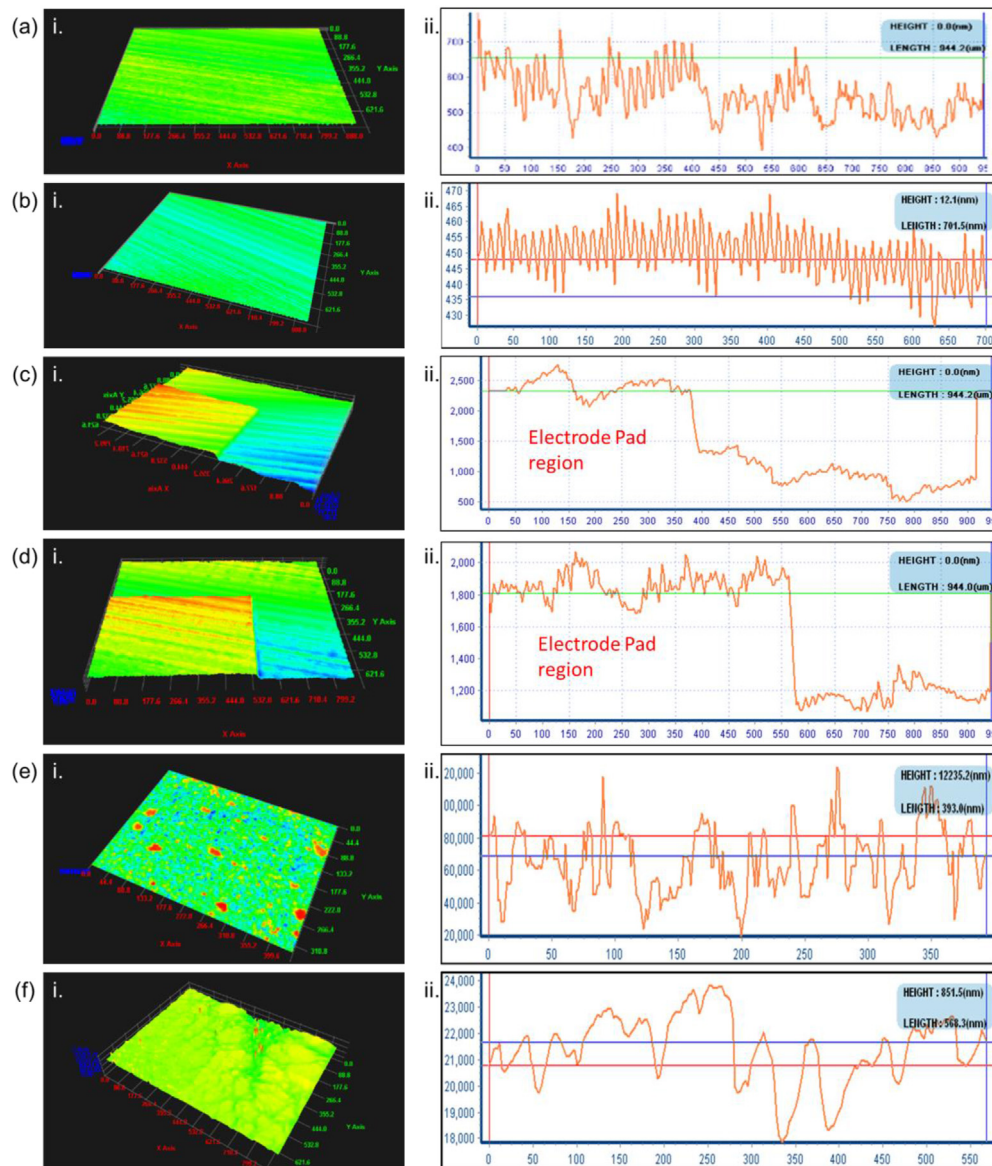


Fig. 2 – 3D images of Si epilayers. 3D surface of (a) Si wafer, (b) oxidized Si wafer, (c) SiO_2 epilayer, (d) single layer of ZnO on SiO_2 , (e) double layer of ZnO on SiO_2 , and (f) ZnO drop casted on SiO_2 . The images in (i) represents the 3D image, whereas (ii) the surface roughness plot generated by 3D profiler.

SiO₂ and ZnO deposited SiO₂ epilayers are smooth and has high potential of semiconductor properties with the accuracy of wet thermal oxidation, photolithography technique and the double layer deposition of ZnO using spin coating technique. Drop casting technique is not preferred due to the undesirable aggregation on semiconductor layers.

3.4. Field emission scanning electron microscope analysis

As the semiconductor epilayers were examined under FESEM, the surface morphology, such surface roughness, and particle distribution were visualized. Fig. 3a shows the electron microscope image of unmodified Si wafer. The insight of BSE image reveals the clear semiconductor surface with invisible impurities. It is supported by the energy dispersive spectroscopy (EDS) data of Si wafer (Fig. 3b), which shows the highest peak for Si and nonexistent carbon (C) and oxygen (O) peaks. The oxidized Si wafer surface, the SiO₂ semiconductor is shown in Fig. 3c. The wet thermal oxidation creates ~50% of oxide layer on the Si substrate. The distinguished black spots observed in the BSE microscopic image of SiO₂ may indicate the presence active oxide layers, which are highly vulnerable to further chemical reactions. It is invalid to notify the black spots as foreign particles as the EDS spectra of SiO₂ epilayer (Fig. 3d) shows highest peak for O and a moderately high peak for Si. The spectra justify the absence of any impurities on the oxidized Si wafer. The small amount of C element appeared might aroused from the sample mounting in the preparation step for FESEM analysis. Fig. 3e shows the morphology of ZnO deposited SiO₂. The presence of sol-gel synthesized nanomaterial on the oxide layer created an unsmooth morphology, showing the deposited wave-like ZnO nanoparticles. A moderately even ZnO nanoparticle distribution is observed from the BSE microscopic image, which emphasizes the spin coating method as the relatively simplest method for nanoparticle deposition on semiconductors. Fig. 3f shows the EDS

spectra of ZnO deposited SiO₂. The elemental spectra justify the effortless preparation of high-performance semiconductor epilayers, as no impurities are shown in the EDS data. Zn, O, and Si peaks are prominent, which verify the developed ZnO/SiO₂ epilayers. The Al component denotes the presence of Al electrode pad on the ZnO deposited SiO₂ for performing electrical analyses.

3.5. Failure analysis at semiconductor epilayers

3.5.1. 3D profiler analysis

Fig. 4 shows the defects found on the fabricated SiO₂ semiconductor epilayer. Fig. 4a(i) shows optical image of a type of defects on the surface. Fig. 4a(ii) reveals rough SiO₂ surface through the fluctuation observed in the surface profile. The 2D image of the defect shown in Fig. 4a(iii) shows assorted accumulation of impurities on the surface, which is classified as surface defect. The 3D vision of the defect observed as in Fig. 4a(iv) and 4a(v) represents the depth of the impurities and patches formed on the SiO₂ surface. Incoherent thermal oxidation may cause such condition to SiO₂ layer, results in the poor performance of the semiconductor. The incoherence refers to the variation of thickness in oxidized Si wafer. Such variation is inherited in the metal deposition, and photolithography process. Thus, the severe variation in thickness appears as patches of incomplete step of process at each stage. Moreover, the patches on SiO₂ surface may occurred due to the left behind photoresist solution during the developing step of fabrication process. When the positive photoresist coated on Al deposited wafer not removed completely during the developing step of fabrication process, the photoresist remains as impurities in the epilayer of SiO₂. Followed by UV exposure and etching process, the possibility to remove the impurities is assumed as null. Thus, the photoresist washing in the fabrication process is crucial to eliminate such defects on the fabricated SiO₂ layer. The photolithography fabrication of interdigitated electrode (IDE) semiconductor is

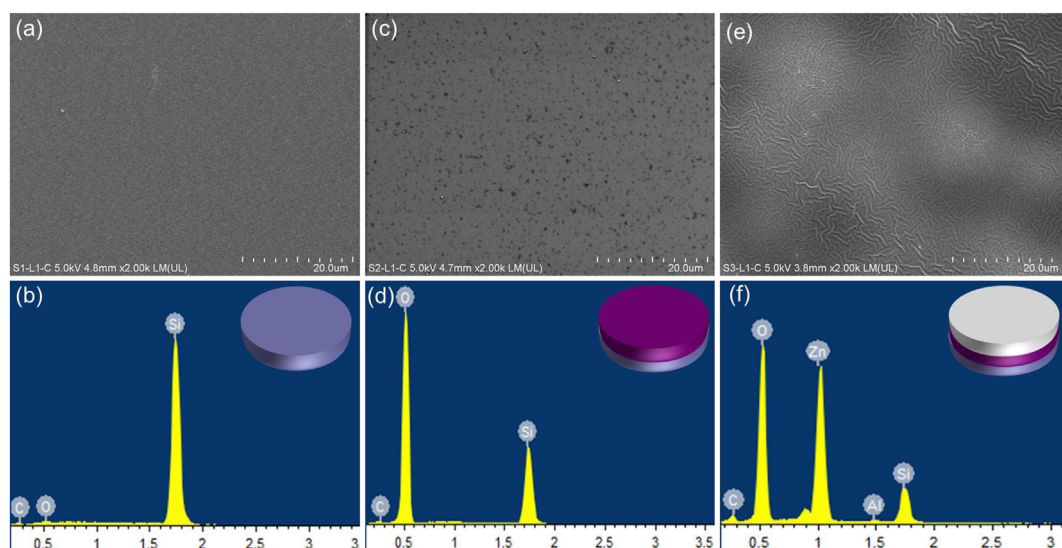


Fig. 3 – SEM analysis of semiconductor epilayers. Electron microscopic image of (a) Si wafer and (b) its EDX spectra. (c) SEM image of SiO₂ epilayer and (d) its EDX spectra. (e) SEM image of ZnO/SiO₂ and its (f) EDX spectra.

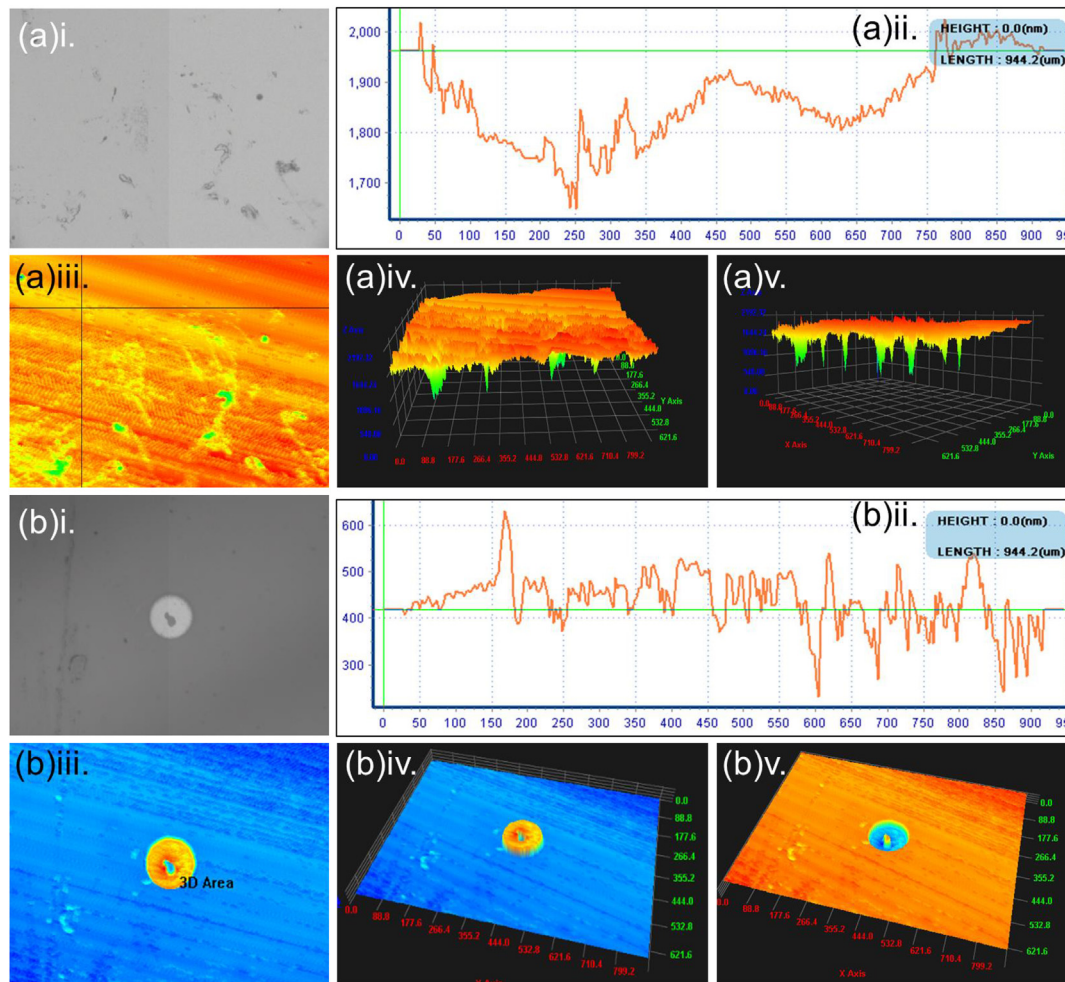


Fig. 4 – Defects on the SiO₂ fabricated semiconductor epilayer. (a) Defect type 1, indicating patches on SiO₂ surface (b) Defect type 2, indicating a circular hole on the SiO₂ surface. i. Optical image of the defect. ii. Surface profiler of the defect. iii. 2D image of defect. iv. 3D image of defect. v. 3D image at different angle.

critical at every step as the defect from one step is carried forward to the next step, which results in an overall malfunction of the device. This is supported based on our previous works and the literature reports on the conventional photolithography fabrication techniques [27–29]. The major cause of defects is related to the man and machine factors. It is important to ensure the machine condition and incorporated materials in a process is at ideal condition. As the semiconductor industry works from the micro-to nano-scale range, a small deviation from the ideal condition may result in the severe surface or line defect. The wet thermal oxidation is performed at high pressure and temperature condition. The even placement of Si wafers and the appropriate settings play is critical to eliminate the defects on SiO₂ surfaces. In many situations, the improper handling of step of photolithography processes causes defects, such incorrect handling of wafers with tweezers, and improper storage condition of oxidized Si wafers. Fig. 4b shows another type of defect found on the SiO₂ surface. A circular hole observed on SiO₂ surface is shown in Fig. 4b(i). The defect is classified as point defect due to the irregularities observed around a single point. The surface roughness at the defect region, shown in Fig. 4b(ii), represents

its uneven surface plane. Moreover, the 2D image (Fig. 4b(iii)) of the defect reveals the apparent size and shape of the defect. The 3D images shown in Fig. 4b(iv) and 4b(v) represent the depth of the point defect on the top and bottom surface. As the circular defect appears sharper and clearer, it is expected as the Al exist on SiO₂ layer. The Al is not completely etched during the etching process. This may be caused by the impurities and error present in the chrome mask, and UV exposure process, which may disrupt a region on the SiO₂ surface. This results in the failure of Al removal in the etching process.

3.5.2. Field emission scanning electron microscope analysis

Fig. 5 shows the FESEM images of defects at each semiconductor epilayers. Fig. 5a shows a type of defect found on the Si wafer. The element present at the defect is shown by the EDX spectra (Fig. 5b), which indicates high peak of carbon. The inserted optical image in Fig. 5b shows the fluorescent view of the surface defect. The carbon may be originated from the delivery systems of wafers. In the absence of clean room condition, organic compounds existed at the materials window, walls, floorings, and ceilings. In those circumstances, the

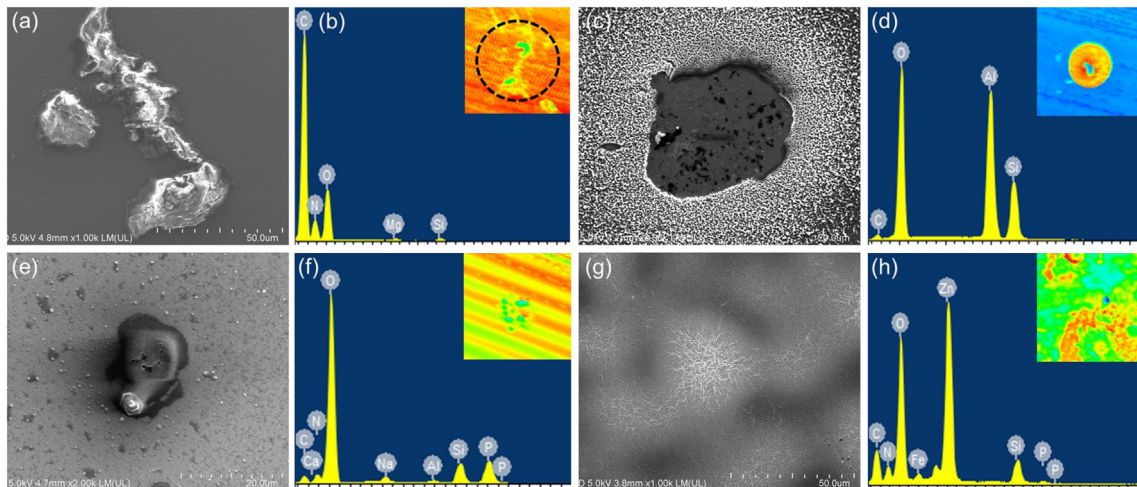


Fig. 5 – SEM images reveal the defects found on semiconductor epilayers. (a) Defect on Si wafer and (b) its EDS spectra. (c) & (e) Defects on SiO₂ epilayer and (d) & (f) its EDS spectra, respectively. (g) Fault found on ZnO/SiO₂ and (h) its EDS spectra. The figure inserted in EDS spectra recalls the fluorescent image of the defects obtained from 3D profiler.

carbon contamination on Si wafer may cause by the wafer storage containers or the tweezers used to hold wafers. The potential sources of carbon contamination can be minimized by storing the wafers at clean room and processing it through UV/ozone cleaning [30]. Fig. 5c reveals defect found on oxidized silicon wafer (SiO₂ epilayer). As identified through 3D profiler analysis (Fig. 4b), the BSE microscopic image indicates the point defect caused by the incomplete Al etching on the SiO₂ epilayer. It is verified by the EDX spectra shown in Fig. 5d, showing high peak of Al and O, and a low Si peak. The inserted figure shows the insight of the defect seen in 3D profiler. The point and surface defect found on SiO₂ semiconductor surfaces may cause severe disruption to the electrical characteristics of the devices. The point or line defects occurs at the electrode pattern of a semiconductor device has high possibility to damage the electric circuit and result in short circuit [31]. The removal of an electrode portion or presence of impurities on the electrodes causes short circuit, due to the undesired components on the electric field. Moreover, the mild surface defects on semiconductor may not cause short circuit, however the device may give substandard electrical readings. The interference in movement of electrons and the charges caused by electrode disruptions results in semiconductor unreliability, depending on the defects [32]. Fig. 5e shows another type of fault found on SiO₂ surface. The defect's elemental composition is shown in Fig. 5f, whereas the inserted image indicates the 3D view of the defect. Based on the elements seen in the EDX spectra, patches found on SiO₂ surface may occurred by the accumulation of impurities on the epilayer, which resulted in the surface defect. A disordered wet thermal oxidation may cause the impurity accumulation as the incomplete active oxide surface is prompt for fast contamination. Fig. 5g shows a clumped and uneven surface of ZnO/SiO₂ epilayer observed under the BSE microscopic image. The EDX spectra of the defect is shown in Fig. 5h, together with the 3D view of the defect (inserted figure). As the 2-layer ZnO was coated on SiO₂ using spin

coater, the uneven distribution of nanomaterial is observed. It is categorized as surface defect, caused by the uneven distribution of ZnO grains. The presence of P, Fe, and N indicate the attachment of contaminating elements on the active ZnO/SiO₂ layer. The smooth ZnO distribution may attained by tuning the rotating speed of spin coater and analyzing the deposition techniques.

3.5.3. Field emission transmission electron microscopic analysis

The FETEM analysis delivers the cross-sectional view of the ZnO/SiO₂ semiconductor epilayer. Fig. 6a(i)-(iv) show the cross-sectional view of ZnO deposited on the SiO₂ epilayer, from low to the high magnifying scale. The cross-sectional analysis of ZnO/SiO₂ semiconductor epilayer shows the line defects observed in a plane of the middle region of the intrinsic semiconductor. Fig. 6a(i) shows the wavy ZnO formed on the SiO₂ surface. As ZnO magnified, the FETEM BF image indicates compacted assembly of spherical shape of ZnO on the semiconductor epilayer. It denotes the uniform size of ZnO synthesized through sol-gel technique. However, the black spots (Fig. 6a(iv)) observed in the ZnO clusters indicate the several impurities present in the epilayer. The contamination may develop during the ZnO deposition or the sample lamella preparation for cross-sectional FETEM analysis. Fig. 6b reveals another type of defect found at the ZnO/SiO₂ epilayer. The uniform and disordered ZnO distribution is shown in Fig. 6b(i), where the patches of air holes are observed between the semiconductor epilayers. Fig. 6b(ii)-(iv) show the magnified BF images of ZnO at the defect area. In comparison to Fig. 6a(iv), the ZnO appeared to be compressed and clumped between the epilayers. Such condition may occur due to hard physical handling of the semiconductor, which has affected the morphology of ZnO/SiO₂ epilayers. Fig. 6c indicates the defected area of ZnO/SiO₂ chosen for lamella preparation. Fig. 6d visualizes the spherical shape ZnO nanoparticle deposited on the semiconductor, observed under FETEM BF

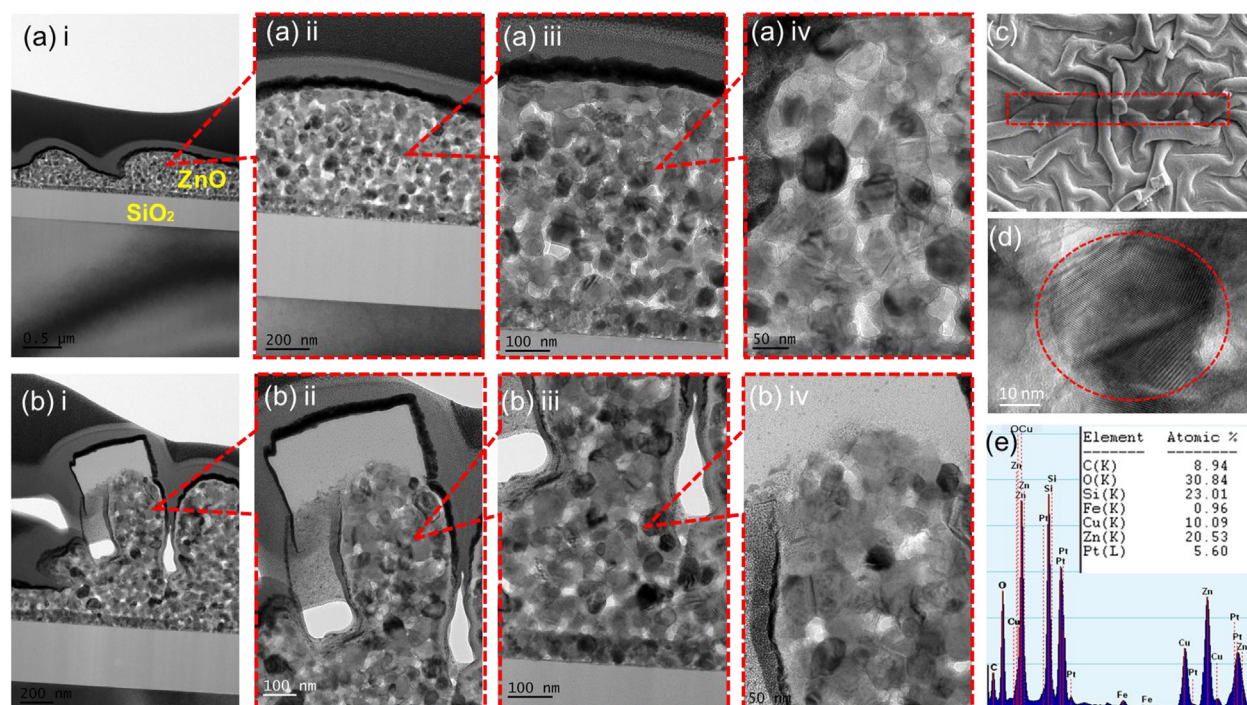


Fig. 6 – FETEM cross-sectional examination of ZnO/SiO₂ defects. (a) i-iv indicate a type of defect found, from low to high scale magnification. (b) i-iv reveal another kind of defect at ZnO/SiO₂ epilayer indicating the clustered assembly of ZnO. (c) Defect area selected to perform lamella preparation for cross sectional analysis. (d) Spherical shaped ZnO nanoparticle captured at the epilayer. (e) EDX spectra of ZnO/SiO₂ epilayer.

images. The elemental composition of the ZnO/SiO₂ defect area is shown in Fig. 6e. The Pt peaks occurred due to the Pt coating of FETEM analysis. Several Cu peaks appeared due to the Cu grid utilized in the scanning processes. The Fe element in the EDX detector is responsible for the small Fe peaks. The high peaks of Zn, Si and O peaks justify the deposition of ZnO on SiO₂ epilayers. The inserted figure shows the atomic percentage of the elements present at the ZnO/SiO₂ epilayer.

4. Conclusion

The research has presented a simple and easy handling photolithography fabrication using Si wafer for developing silicon (Si) semiconductor epilayers. The study has emphasized the simplicity of lab-based photolithography technique for fabricating precise and accurate Si semiconductor and its epilayer for developing nanoscale dimensioned electronic components. Optical and electron microscopic analyses justified the fabrication precision and Si semiconductor accuracy at laboratory facility. The classic incorporation of nanoscale metal oxides with Si semiconductors were presented using sol-gel synthesized zinc oxide (ZnO) to indicate the visualization of nanomaterial integrated Si semiconductor epilayers through the microscopic analysis. The precision of Si epilayers integrated with ZnO through spin coating technique were justified through the high-power microscope (HPM), 3D profilometer, field emission scanning electron microscope (FESEM), and field emission transmission electron microscope

(FETEM) analyses, which revealed the morphology and the deviations from the ideal semiconductor epilayer. The deviations were inspected and studied to determine the root cause of defect using microscopic analysis. The study justified the lab-based failure analysis of Si semiconductor epilayers with the nanomaterial modifications for futuristic development of defect-free semiconductors with easy and inexpensive lab-based techniques. Although modern defect analysis methods are proposed and implemented, the electron and optical microscopic analysis may serve as the fundamental failure analysis tool for fabricating Si semiconductor epilayers by excluding the basic complications at the laboratory facility, prior to the large-scale semiconductor industry.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jmrt.2022.10.116>.

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